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GST 103: Philosophy and Logic - Arguments, Fallacies and Clear Thinking

Learn to tell a real argument from noise, and to build one that survives cross-examination

GST / COMPULSORY

GST & Compulsory Courses

General Studies (GST / CED)
GST Directorate | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Why this pack exists

Philosophy and Logic is where many science and management students switch off - until they meet the objective questions. This pack explains validity, soundness, syllogisms and informal fallacies in plain language, with everyday UNIZIK examples and a fallacy-spotting drill.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Distinguish an argument from a statement, explanation or quarrel.
- Test an argument for validity and soundness without confusing the two.
- Recognise the major informal fallacies in speeches, adverts and social-media arguments.
- Translate simple arguments into standard form and evaluate categorical syllogisms.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

What philosophy is doing

Philosophy is not a collection of opinions. It is the disciplined practice of asking what a claim means, what it assumes, and what would count as evidence for it.

The main branches you will meet are metaphysics (what exists), epistemology (what knowledge is and how we get it), ethics (how we should act), aesthetics (beauty and art) and logic (the rules of correct reasoning).

For examinations, logic carries the most marks because it is teachable and testable. The other branches matter for definitions and short-answer questions.

- Metaphysics: reality, mind, causation, freedom.

- Epistemology: belief, truth, justification, scepticism.
- Ethics: right action, virtue, consequences, duty.
- Logic: arguments, validity, proof, fallacy.

Arguments: the basic unit

An argument is a set of statements where some, the premises, are offered as support for another, the conclusion.

'I am tired, therefore I will sleep' is an argument. 'I am tired because I slept late' is an explanation. 'You are wrong because you are annoying' is an insult wearing an argument's clothes.

Premises answer 'what is the evidence?' The conclusion answers 'what follows?' Indicator words help: therefore, hence, so, thus and consequently usually introduce conclusions; because, since, given that and for usually introduce premises.

> **Validity versus soundness**
> Validity is about structure: if the premises were true, the conclusion would have to be true. Soundness is validity plus actually true premises. An argument can be valid and unsound, but never sound and invalid.

Three patterns worth memorising

Most exam syllogisms use a small number of forms. Learn the patterns and you can evaluate them quickly.

- Modus ponens: If P then Q. P. Therefore Q. Valid.
- Modus tollens: If P then Q. Not Q. Therefore not P. Valid.
- Affirming the consequent: If P then Q. Q. Therefore P. Invalid - other causes of Q may exist.
- Denying the antecedent: If P then Q. Not P. Therefore not Q. Invalid - Q may still happen for another reason.
- Disjunctive syllogism: Either P or Q. Not P. Therefore Q. Valid when the options are genuinely exclusive.

Categorical syllogisms in plain language

A categorical syllogism has two premises and a conclusion built from three terms: major, minor and middle.

Example: All students write examinations. Ada is a student. Therefore Ada writes examinations. The middle term, 'students', links the other two terms.

Watch the quantifiers: all, no, some, some not. 'Some' means at least one, not most. A conclusion cannot introduce a stronger claim than the premises support.

- All A are B; all B are C; therefore all A are C. Valid.
- No A are B; some C are A; therefore some C are not B. Valid.
- All A are B; some B are C; therefore some A are C. Invalid - the C may be among the Bs that are not A.

Informal fallacies: the ones you will actually meet

A fallacy is a pattern of reasoning that looks persuasive but fails. Learning the names is useful; learning to explain why each fails is what earns marks.

- Ad hominem: attacking the person instead of the argument.
- Appeal to authority: citing a famous person outside their expertise.
- Appeal to popularity: 'everybody does it' does not make it true or right.
- False dilemma: presenting two options when more exist.
- Slippery slope: claiming one small step must lead to an extreme outcome without evidence.
- Straw man: attacking a weaker version of an opponent's position.
- Circular reasoning: assuming the conclusion in the premises.
- Hasty generalisation: concluding from a tiny or biased sample.
- Post hoc: assuming that because B followed A, A caused B.

> **In the exam**
> When asked to identify a fallacy, name it and write one sentence explaining the reasoning error. A name without an explanation is half an answer.

Validity is not truth

Logic studies the relation between premises and conclusion, not the truth of the premises themselves. An argument is valid when the conclusion cannot be false if the premises are true. It is sound when it is valid and the premises are also true. These are separate questions, and confusing them is the most common error in first-year logic.

"All lecturers are mortal; Dr Ade is a lecturer; therefore Dr Ade is mortal" is valid and, as it happens, sound. "All lecturers are wealthy; Dr Ade is a lecturer; therefore Dr Ade is wealthy" is equally valid - the form is identical - but unsound, because the first premise is false. Logic cannot tell you whether lecturers are wealthy; it can tell you that if the premise were true the conclusion would follow.

This is why logic is useful: it lets you accept a conclusion provisionally while disputing a premise, and it exposes arguments that look persuasive because of their rhetoric rather than their structure.

The forms you must recognise

Modus ponens: if P then Q; P; therefore Q. Valid. Modus tollens: if P then Q; not Q; therefore not P. Valid. Denying the antecedent: if P then Q; not P; therefore not Q. Invalid. Affirming the consequent: if P then Q; Q; therefore P. Invalid. These four forms account for a large share of examination questions.

Categorical syllogisms work through quantity and quality: All S are P; some S are M; therefore some M are P. The techniques of distribution and the Venn diagram test are used to check validity without relying on intuition.

Informal fallacies are failures of relevance or evidence rather than form: ad hominem (attacking the person), appeal to authority (citing status instead of evidence), false dilemma (two options when more exist), begging the question (assuming the conclusion), straw man (misrepresenting an opponent), and post hoc (assuming sequence proves causation).

> **Name the fallacy precisely**
> In an exam, "this is a fallacy" earns nothing. Identify the form or name the fallacy and say exactly which move is illegitimate.

Philosophy as the examination of assumptions

The philosophical questions in this course - what is knowledge, what is real, what makes an action right - are asked because ordinary assumptions conflict when pressed. Knowledge as justified true belief faces the problem of accidentally justified belief: a person who believes the time by looking at a stopped clock that happens to show the right time has a true justified belief but does not know the time. The discussion of such cases is the method, not an interruption of it.

Branches to keep distinct: epistemology (knowledge), metaphysics (what exists), ethics (how to act), logic (correct inference), aesthetics (beauty and art). A good answer places a question in its branch before attempting it, because the standards of argument differ.

Write philosophy in the first person plural of argument: "One might object that..." then answer. Assertion without argument is the characteristic failure, and it is easy to detect: if your paragraph could be prefaced by "I just think that", it is not yet philosophy.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. Testing an argument for validity

A newspaper argues: "If the policy were sound, enrolment would have risen. Enrolment has risen. Therefore the policy is sound."

1. Formalise: if P then Q; Q; therefore P.
2. Name the form: affirming the consequent.
3. Show why it fails: enrolment could rise for other reasons.
4. Construct a parallel case with an obviously false conclusion to make the failure vivid.
5. State what would be needed: evidence that no other factor could explain the rise.

> **What to carry forward**
> Formalise first, name second, explain third. Intuition alone will not earn the marks.

2. Answering an epistemology question

"Is justified true belief sufficient for knowledge?" Answer in fifteen minutes.

1. Define the three conditions and the classical analysis.
2. State the challenge with the stopped-clock or barn-façade case.
3. Explain precisely what the case shows: justification is accidentally connected to truth.
4. Consider one response: add a no-false-lemmas or reliability condition.
5. Assess the response: does it exclude the case without excluding ordinary knowledge?
6. Conclude with a position and the cost of holding it.

> **What to carry forward**
> Philosophy answers have the shape: thesis, counter-example, reply, assessment, position.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. Distinguish validity from soundness and explain why the distinction matters.

15 marks

1. Define validity as form: premises cannot be true with a false conclusion.
2. Define soundness as validity plus true premises.
3. Give a valid but unsound example.
4. Explain the practical use: allows criticism of premises independently of inference.

Q2. Identify the fallacy: "You cannot criticise the new timetable; you are not a member of the senate."

10 marks

1. Name the move: rejecting a criticism because of the critic's status.
2. Identify the fallacy: ad hominem, or appeal to authority in reverse.
3. Explain why it fails: the truth of a criticism does not depend on who makes it.
4. Note the legitimate version: expertise may affect the weight of testimony, not the validity of an argument.

Q3. Is knowledge more than justified true belief? Discuss.

20 marks

1. Set out the tripartite analysis.
2. Present a Gettier-style counter-example in detail.
3. Explain the intuition the case produces.
4. Evaluate at least two proposed additions to the analysis.
5. Conclude with a defended position and its cost.

> **Where marks are lost**
> Describing the counter-example without saying what it is supposed to show.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Validity. A property of form: true premises cannot yield a false conclusion.

Soundness. Validity combined with actually true premises.

Modus ponens. If P then Q; P; therefore Q - a valid form.

Modus tollens. If P then Q; not Q; therefore not P - a valid form.

Affirming the consequent. If P then Q; Q; therefore P - an invalid form.

Denying the antecedent. If P then Q; not P; therefore not Q - an invalid form.

Syllogism. A deductive argument with two premises and a conclusion.

Ad hominem. Rejecting a claim by attacking the person who made it.

False dilemma. Presenting two options as exhaustive when others exist.

Begging the question. Assuming the conclusion among the premises.

Epistemology. The branch of philosophy concerned with knowledge and justification.

Counter-example. A case showing that a general claim or analysis fails.

Your four-week plan

Week 1 - Form

Validity.

- Formalise ten short arguments.
- Test each for validity.
- Write one valid but unsound argument deliberately.

Week 2 - Syllogisms

Categorical logic.

- Translate sentences into All/Some/No form.
- Test five syllogisms with Venn diagrams.
- Identify the distributed terms.

Week 3 - Fallacies

Naming and explaining.

- Collect five fallacies from news commentary.
- Name each and state the illegitimate move.
- Write a corrected version of one.

Week 4 - Philosophy

Essay shape.

- Outline a thesis, counter-example and reply.
- Write one full answer under time.
- Check that every claim has an argument.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any introductory logic text covering validity, syllogisms and informal fallacies.
- An introduction to philosophy with primary extracts on knowledge and ethics.
- A dictionary of philosophy for precise definitions of terms.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. Is this valid or sound: 'All birds fly. Penguins are birds. Therefore penguins fly.'?

Answer: The form is valid, but the argument is unsound because the first premise is false. Soundness requires true premises as well as valid structure.

2. Identify the fallacy: 'Don't listen to his advice on saving money - he failed his first business.'

Answer: Ad hominem. The speaker attacks the person rather than addressing the financial advice.

3. What is the difference between an explanation and an argument?

Answer: An argument offers evidence for a conclusion. An explanation shows why something is the case, assuming the fact is already accepted.

4. Test the form: If it rains, the ground is wet. The ground is wet. Therefore it rained.

Answer: Invalid - affirming the consequent. The ground could be wet for another reason, such as a burst pipe.

5. Name the fallacy: 'Either you support the strike or you hate students.'

Answer: False dilemma. It presents only two options when other positions are possible.

6. Why is 'some' different from 'most' in logic?

Answer: 'Some' means at least one. 'Most' means more than half. A conclusion using 'most' cannot be inferred from premises that only claim 'some'.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write definitions of validity and soundness from memory, each with one example.
- [ ] Build your own valid modus ponens and invalid affirming-the-consequent argument on a campus topic.
- [ ] Collect ten headlines or adverts and name the fallacy in at least five.
- [ ] Draw a table of the six fallacies you find hardest and write a one-line rebuttal for each.
- [ ] Take a past objective question on logic and explain why each wrong option is wrong.
- [ ] Have a friend give you a one-paragraph argument and evaluate its premises and conclusion.
- [ ] Teach the difference between validity and soundness to someone in five minutes.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date: